

Multiple Choice

1. **Answer:** D

Options: A) Nitrogen; B) Argon; C) Methane; D) CO₂

Note: Correct option: D - CO₂

2. **Answer:** C

Options: A) A kilogram of feathers; B) Five pounds of bricks as measured on Earth; C) Five kilograms of feathers; D) A kilogram of bricks

Note: Correct option: C - Five kilograms of feathers

3. **Answer:** C

Options: A) The first time you drop a bowling ball and it falls to the ground proving your hypothesis.; B) After you've repeated your experiment many times.; C) You can never prove your theory to be correct only "yet to be proven wrong".; D) When you and many others have tested the hypothesis.

Note: Correct option: C - You can never prove your theory to be correct only "yet to be proven wrong".

4. **Answer:** B

Options: A) 130 million km; B) 150 million km; C) 170 million km; D) 190 million km

Note: Correct option: B - 150 million km

5. **Answer:** C

Options: A) 240 to 680 nm.; B) 360 to 620 nm.; C) 380 to 740 nm.; D) 420 to 810 nm.

Note: Correct option: C - 380 to 740 nm.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '9:4'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** the presence of life on Earth. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** About 6 meters. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key